

6.2 Causes and effects of the digital divide

The digital divide describes the gap between people who have the latest technology and those that do not. There are many causes of the digital divide.

Access to electricity

Technology requires electricity. Devices can run on battery or directly from mains electricity. Even if a device uses a battery, it still needs to be recharged.

It can be difficult to install electrical cables in remote parts of the world. It is expensive to build cables and install them in places many hundreds of miles away from power stations. Examples of places like these include deserts, mountains and remote villages or towns.

Solar power can help reduce the impact of being in a remote place. However, solar power is expensive to install.

Without access to electricity, people cannot use digital devices effectively.

Cost of technology

New technology is very expensive. Old technology is cheaper than the newest technology but will not be able to perform the same functions, access new programs or be as reliable. People from poorer areas of the world may not be able to afford new technology, or they may not be able to afford electricity to power the technology. They may not be able to pay for repairs if the technology breaks.

The cost of technology can create a digital divide in big cities. Cities have shops that sell the new technology. But if people have low paid jobs, they may not be able to afford to buy or rent new technology.

Ability to repair devices

Digital devices often need repairing or upgrading. Upgrades could include adding storage or updating software. If a person is unable to find a way to repair their device, for example by a local computer or phone technician, they will not be able to use their digital device (see Figure 6.3). A person with a particular brand of mobile device may not be able to get their device repaired locally. This may mean they need to buy a new device.



Figure 6.3: One cause of the digital divide is a lack of easy access to repair devices.

Access to the internet and wireless connections

Access to the internet is important as many services are only available by using the internet. For example, video conferencing, email/messaging apps and websites all require internet access. However, certain things like using word processing apps and printing may not need the internet.

Remote places make it difficult to install internet connections. The use of mobile phone networks and satellites removes the need for physical connections. However, satellite connections cost a lot of money to use. Satellite connections can also be slow and unreliable. Because the signals travel through the atmosphere into space, and back again, they can be disrupted by weather conditions like thunderstorms. Mobile phone networks only have a limited range and may not reach remote locations.

Very slow connections to the internet, or a total lack of connection adds to the digital divide. Lack of internet is probably one of the biggest factors when discussing the digital divide.

Availability of technology

The availability of technology refers to how many digital devices there are compared with how many people want them. For example, there may be 10 000 people who can afford a digital device, have access to the internet and are able to use these devices. However, if there are only 5000 devices available, then 5000 people will still not be able to access a digital device.

The COVID-19 pandemic meant that manufacturing of many things was disrupted. Prices of digital devices increased significantly at this time. This is because the microchips

needed to manufacture these devices were not produced in the numbers that were needed.

New technology may need new manufacturing processes or better robots to build. This may limit the quantity of high-performance technology that is available. This increases the price for the technology.